AKASHDIP MAHAPATRA | TECHNICAL PORTFOLIO ADDENDUM

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ENTERPRISE SYSTEM ARCHITECTURE

Project: ODIE Serverless Data Pipeline (Aerospace IoT & Telemetry)

Role: Data Engineer / DevOps & Infrastructure Maintainer

Scale: Production-grade, high-throughput, multi-tiered AWS serverless architecture processing continuous real-time aviation telemetry and IoT data.

System Architecture Overview:

The system is a decoupled, event-driven data lake consisting of distinct ingestion, processing, storage, and distribution layers.

- **Ingestion: High throughput** streaming via Amazon MSK (Kafka) and AWS IoT Core.
- **Routing:** Heavy utilization of Amazon **EventBridge** to trigger asynchronous workflows based on object creation in S3 Raw Zones.
- **Compute:** Python-based transformation logic running in containerized AWS Lambda functions (managed via ECR).
- **Storage & Distribution:** Tiered S3 Data Lake (Raw, Processed, Curated), DynamoDB for real-time state management, and Amazon SNS / Kinesis Firehose for downstream data distribution.

Key Engineering Responsibilities:

- **CI/CD Pipeline Ownership:** Maintain highly structured GitHub Actions workflows (lambda-container-build-dev.yaml, terraform-release.yaml) handling the end-to-end build, testing, and deployment of Dockerized Lambda functions and Terraform modules across Dev, QA, UAT, and Prod environments.
- **Infrastructure as Code (IaC):** Manage complex infrastructure using modular Terraform, defining VPCs, IAM roles with strict least-privilege policies, S3 event triggers, and Lambda resource provisioning.
- **Container Security & Maintenance:** Monitor Dockerfiles and requirements.txt dependencies across multiple microservices (e.g., *AIDX Sequencer, Aircraft Towing Processor, GSE Tracking*), actively identifying and patching CVEs to maintain strict enterprise security standards.
- **Observability & Troubleshooting:** Utilize AWS CloudWatch and Datadog for system-wide tracing, log aggregation, and error handling (managing Dead Letter Queues) to ensure zero data loss in the ingestion layer.

Hardware & Academic Engineering

APPLIED NETWORKING & IOT ENGINEERING (Selected Projects)

1. Network Attached Storage (NAS) & Custom Web Hosting via ESP32

- **Tech Stack:** ESP32, Arduino, HTTP, Local Area Network (LAN).
- **Details:** Engineered a microcontroller-based NAS to serve files over Wi-Fi. Extended this architecture to host a custom HTML web interface directly on the ESP32 for localized hardware control, simulating a micro web server environment.

2. Wi-Fi Penetration & Network Security Tool

- **Tech Stack:** ESP8266/ESP32, Python, Packet Sniffing.
- **Details:** Designed a wireless network testing system capable of scanning 802.11 traffic, capturing handshake packets, and broadcasting de-authentication packets to test network access point vulnerabilities.

3. Real-Time GPS Tracking & Telemetry System

- **Tech Stack:** Node.js, Express, Socket.IO, ESP32, GPS Modules.
- **Details:** Built an end-to-end IoT telemetry pipeline. Captured live GPS coordinates on edge hardware (ESP32) and streamed the data via WebSockets to a Node.js backend for real-time mobile visualization.

4. Lightweight Python Web Server (From Scratch)

- **Tech Stack:** Python, TCP Sockets, OSI Model Layer 4/7.
- **Details:** Bypassed standard frameworks (like Flask/Django) to build a raw HTTP server using basic Python TCP socket programming. Handled low-level GET/POST request parsing and static content delivery to solidify core networking fundamentals.

PROFESSIONAL CREDENTIALS & CERTIFICATIONS

Verify in Credly - https://www.credly.com/users/akashdip2001/

GitHub Advanced Security

ISSUED TO

AKASHDIP MAHAPATRA

Issued on: 10 FEB 2025 | Expires on: 10 FEB 2028 | Issued by: GitHub

Verify: https://www.credly.com/go/0uS9HRCw

Microsoft

AKASHDIP MAHAPATRA

has successfully passed all requirements for

Microsoft Certified: Azure Network Engineer Associate

Credential ID: AA9E4130816324D7

Certification number: D498AD-9345B3

Earned on: 20 October 2025

Expires on: 21 October 2026

Online Verifiable

Microsoft CERTIFIED ASSOCIATE

Satya Narayana Nadella

aws certified

AKASHDIP MAHAPATRA

AWS Certified Cloud Practitioner

VALIDATION NUMBER: 5cd5b33e061a49758219e5c946327796

VALIDATE AT: https://aws.amazon.com/verification

Issue Date: June 22, 2025

Expiration Date: June 22, 2028